Delta Electronics, Inc.
PSS®E Dynamic Model for
Delta PCS3440s

Contents

Executive Summary

This report describes the PSSE user-defined model developed by Siemens PTI for the Delta PCS3440s. The following PSSE User-defined models were used to represent the dynamic behavior of Delta PCS3440s for transient stability analysis.

a) DELTAESS_GC – User-defined generator/converter model,

b) DELTAESS_PC– User-defined plant control model.

A series of benchmark tests were performed to validate the developed model.

i. Flat-run test,

ii. Power rising test,

iii. P setpoint response test,

iv. Q setpoint response test,

v. Protection function tests, and

vi. Fault ride-through tests.

The benchmark test results indicate that the developed PSSE model exhibits reasonable and accurate responses within the context of establishing PSSE dynamic model. The provided information offers crucial insights into the system response.

1. Introduction

For the purpose of conducting a regulatory-compliant transient response analysis, a set of PSSE dynamic models were developed to represent the dynamic behavior of the Delta PCS3440s converter, serving as a foundational framework for in-depth investigation and analysis within this report.

The PSS®E User-defined models **DELTAESS_GC**, and **DELTAESS_PC** is used for the control functions and converter dynamics of the PCS3440s.

This report describes the parameter, benchmark tests and final simulation models.

The rest of this document is organized as follows:

- The model description, including the test system, are presented in Section 2 of this report.
- In Section 3, the parameters of the dynamic models are described.
- The benchmark tests together with the validation results are presented in Section 4 of this document.
- Finally, the discussions, conclusions and recommendations from this work are provided in Section 5.

2. Model Description and Test System

This section describes the Delta PCS3440s model and its test system used in the PSS®E simulation model.

2.1 The Test System

The Test System shown in Figure 1 contains the following circuit components:

- Utility equivalent
- PCS3440s

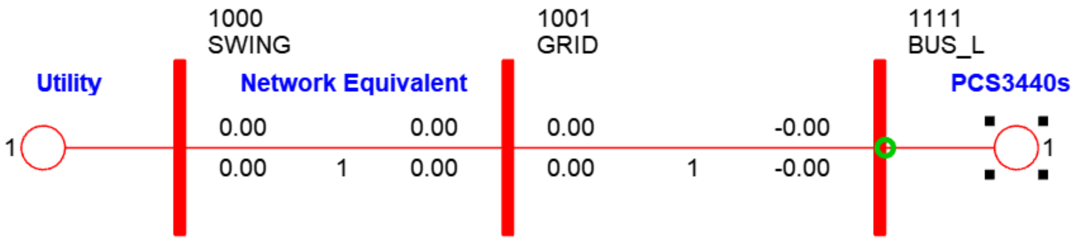

Figure 1. Single Line Diagram of the Test System

2.2 Utility Equivalent

Utility equivalent is modeled as 0.69kV equivalent generator with one branch.

- The equivalent generator is connected to the Swing Bus 1000. It is modeled using PSS®E standard library model **PLBVF1.**
- The branch is connected between bus bars 1000 to 1001.

2.3 Delta PCS3440s

The Delta PCS3440s uses PSS®E renewable generator power flow model, as shown in Figure 2.

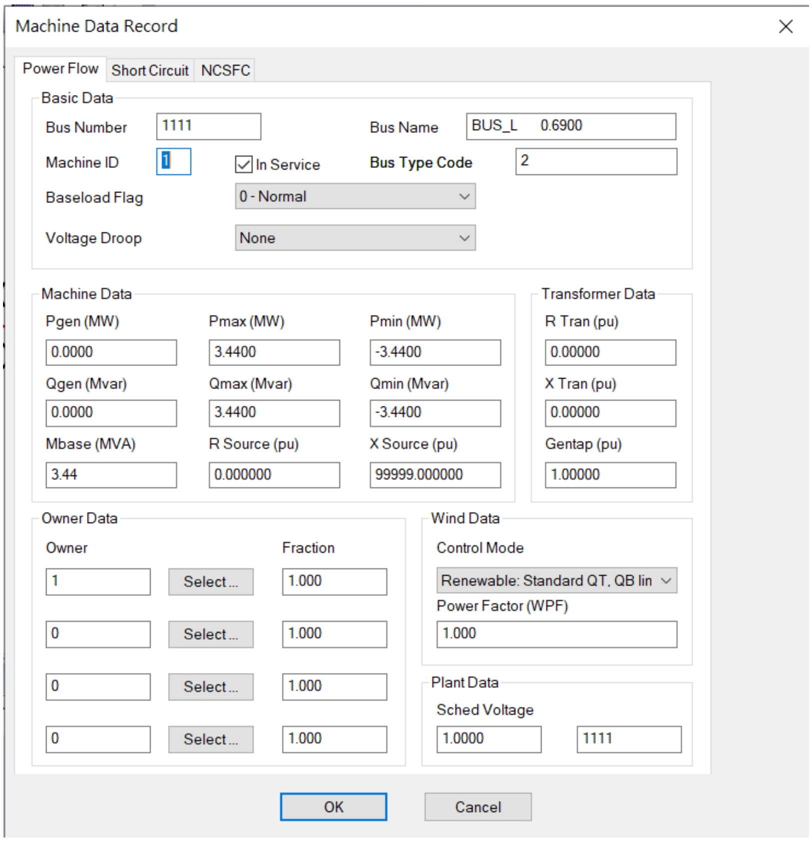

Figure 2. Delta PCS3440s Renewable Generator Power Flow

3. PSS®E Dynamic Models of the PCS3440s

This section mainly introduces the models developed for the Delta PCS3440s in PSS®E dynamic simulation. The PCS3440s modeling is composed of two models including **DELTAESS_PC** and **DELTAESS_GC**, as shown in Figure 3.

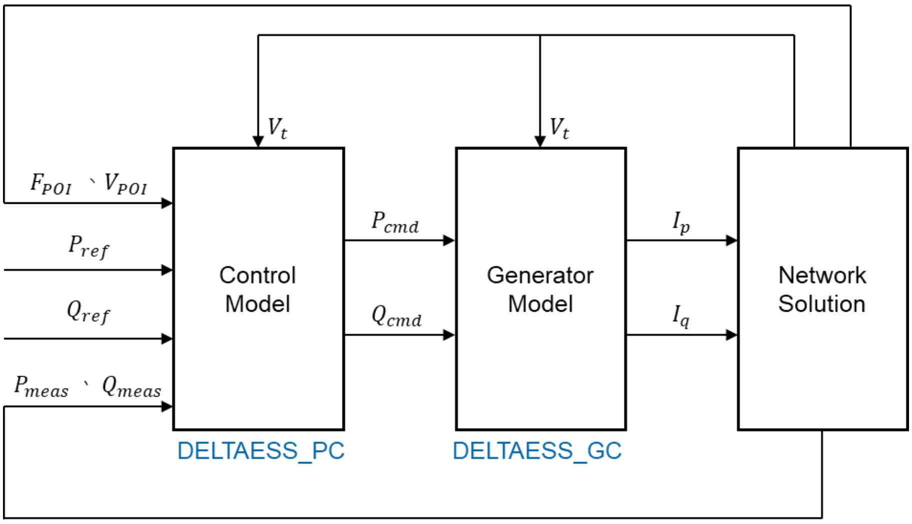

Figure 3. Delta PCS3440s modeling architecture

3.1 Generator Model – DELTAESS_GC

The DELTAESS_GC model process real and reactive power commands from the plant control module, with feedback of terminal voltage for lower voltage active current and high voltage reactive current management logics, and outputs real and reactive current injections into the network model.

The parameters of the model are shown below.

Table 1. Integer Constants Parameters for DELTAESS_GC Model

ICONs	Value	Description
M	2	ILIM_MODE, 0=vector, 1=P-priority, 2=Q-priority
M+1	1	FRT_MODE, 0=disable K-factor, 1=enable K-factor

Table 2. Real Constants Parameters for DELTAESS_GC Model

CONs	Value	Description
J	1.2000	IMAX, current limit on machine base
J+1	0.2000	VMIN_INJ, minimum voltage for current calculation
J+2	1.0000	VREF, voltage reference for FRT K-factor
J+3	2.0000	KQF, K-factor for reactive current compensation
J+4	0.1000	DBDL, low-voltage deadband
J+5	0.1000	DBDH, high-voltage deadband
J+6	0.6000	IQFMAX, maximum FRT Iq on machine base
J+7	-0.6000	IQFMIN, minimum FRT Iq on machine base
J+8	1.0000	PMAX, maximum Pref on machine base
J+9	-1.0000	PMIN, minimum Pref on machine base
J+10	0.5000	QMAX, maximum Qref on machine base
J+11	-0.5000	QMIN, minimum Qref on machine base

3.2 Plant Controller model – DELTAESS_PC

The plant controller processes frequency and active power output to emulate frequency/active power control. It also emulates volt/var control at the plant level. This model provides active and reactive power commands to the electrical control model. The parameters of the model are shown below.

Table 3. Integer Constants Parameters for DELTAESS_PC Model

ICONs	Value	Description
M	1111	Remote bus number or 0 for local voltage control
M+1	0	EN_PR_GC_NORM_RR_EN (0:Disable; 1:0.01%; 2:0.1%; 3:1%)
M+2	127	EN_PR_GC_AC_VOLT_LEVEL_EN (0:Disable; 1:Enable)
M+3	56	EN_PR_GC_AC_FREQ_LEVEL_EN (0:Disable; 1:Enable)
M+4	0	EN_PR_GC_VVAR_MODE_EN (0:Disable; 1:Enable)
M+5	0	EN_PR_GC_VWATT_MODE_EN (0:Disable; 1:Enable)
M+6	0	EN_PR_GC_FWATT_OF_MODE_EN (0:Disable; 1:Enable)
M+7	0	EN_PR_GC_FWATT_UF_MODE_EN (0:Disable; 1:Enable)
M+8	0	EN_PR_GC_WATTVAR_MODE_EN (0:Disable; 1:Enable)

Table 4. Real Constants Parameters for DELTAESS_PC Model

CONs	Value	Description
J	6900.0000	EN_PR_GC_RATING_AC_VOLT (0.1V)
J+1	6000.0000	EN_PR_GC_RATING_AC_FREQ (0.01Hz)
J+2	34400.0000	CMD_ACTIVE_POWER_DEMAND (0.1kW)
J+3	0.0000	CMD_REACTIVE_POWER_DEMAND (0.1kVAR)
J+4	33.0000	EN_PR_GC_NORM_RR_VAL (0.01/0.1/1%)
J+5	1200.0000	EN_PR_GC_AC_OVP_HV4 (0.1%)
J+6	1200.0000	EN_PR_GC_AC_OVP_HV3 (0.1%)
J+7	1100.0000	EN_PR_GC_AC_OVP_HV2 (0.1%)
J+8	1050.0000	EN_PR_GC_AC_OVP_HV1 (0.1%)
J+9	900.0000	EN_PR_GC_AC_UVP_LV1 (0.1%)
J+10	700.0000	EN_PR_GC_AC_UVP_LV2 (0.1%)
J+11	500.0000	EN_PR_GC_AC_UVP_LV3 (0.1%)
J+12	250.0000	EN_PR_GC_AC_UVP_LV4 (0.1%)
J+13	0.0000	EN_PR_GC_AC_OVP_TIME_HV4 (0.01s)
J+14	0.0000	EN_PR_GC_AC_OVP_TIME_HV3 (0.01s)
J+15	100.0000	EN_PR_GC_AC_OVP_TIME_HV2 (0.01s)
J+16	180000.0000	EN_PR_GC_AC_OVP_TIME_HV1 (0.01s)
J+17	600.0000	EN_PR_GC_AC_UVP_TIME_LV1 (0.01s)
J+18	300.0000	EN_PR_GC_AC_UVP_TIME_LV2 (0.01s)
J+19	120.0000	EN_PR_GC_AC_UVP_TIME_LV3 (0.01s)
J+20	32.0000	EN_PR_GC_AC_UVP_TIME_LV4 (0.01s)
J+21	6180.0000	EN_PR_GC_AC_OFP_HF3 (0.01Hz)
J+22	6180.0000	EN_PR_GC_AC_OFP_HF2 (0.01Hz)
J+23	6120.0000	EN_PR_GC_AC_OFP_HF1 (0.01Hz)
J+24	5880.0000	EN_PR_GC_AC_UFP_LF1 (0.01Hz)
J+25	5700.0000	EN_PR_GC_AC_UFP_LF2 (0.01Hz)
J+26	5700.0000	EN_PR_GC_AC_UFP_LF3 (0.01Hz)
J+27	5700.0000	EN_PR_GC_AC_UFP_LF4 (0.01Hz)
J+28	10.0000	EN_PR_GC_AC_OFP_TIME_HF3 (0.01s)
J+29	29900.0000	EN_PR_GC_AC_OFP_TIME_HF2 (0.01s)
J+30	2990.0000	EN_PR_GC_AC_OFP_TIME_HF1 (0.1s)
J+31	2990.0000	EN_PR_GC_AC_UFP_TIME_LF1 (0.1s)
J+32	10.0000	EN_PR_GC_AC_UFP_TIME_LF2 (0.01s)
J+33	10.0000	EN_PR_GC_AC_UFP_TIME_LF3 (0.01s)
J+34	10.0000	EN_PR_GC_AC_UFP_TIME_LF4 (0.01s)
J+35	80.0000	EN_PR_GC_VVAR_V1 (0.1%)
J+36	149.7000	EN_PR_GC_VVAR_Q1 (0.1%)
J+37	20.0000	EN_PR_GC_VVAR_V2 (0.1%)
J+38	23.9500	EN_PR_GC_VVAR_Q2 (0.1%)

CONs	Value	Description
J+39	20.0000	EN_PR_GC_VVAR_V3 (0.1%)
J+40	23.9500	EN_PR_GC_VVAR_Q3 (0.1%)
J+41	80.0000	EN_PR_GC_VVAR_V4 (0.1%)
J+42	-149.7000	EN_PR_GC_VVAR_Q4 (0.1%)
J+43	0.0000	EN_PR_GC_VVAR_RESPON_TIME (0:Immediately; 1:0.01s)
J+44	1000.0000	EN_PR_GC_VVAR_VREF (0.1%)
J+45	900.0000	EN_PR_GC_VWATT_V1 (0.1%)
J+46	1000.0000	EN_PR_GC_VWATT_P1 (0.1%)
J+47	950.0000	EN_PR_GC_VWATT_V2 (0.1%)
J+48	1000.0000	EN_PR_GC_VWATT_P2 (0.1%)
J+49	1060.0000	EN_PR_GC_VWATT_V3 (0.1%)
J+50	1000.0000	EN_PR_GC_VWATT_P3 (0.1%)
J+51	1100.0000	EN_PR_GC_VWATT_V4 (0.1%)
J+52	-1000.0000	EN_PR_GC_VWATT_P4 (0.1%)
J+53	0.0000	EN_PR_GC_VWATT_RESPON_TIME (0:Immediately; 1:0.01s)
J+54	3.0000	EN_PR_GC_FWATT_OF_DEADBAND (0.1%)
J+55	3.0000	EN_PR_GC_FWATT_UF_DEADBAND (0.1%)
J+56	50.0000	EN_PR_GC_FWATT_OF_DROOP_RATIO (0.1%)
J+57	50.0000	EN_PR_GC_FWATT_UF_DROOP_RATIO (0.1%)
J+58	0.0000	EN_PR_GC_FWATT_RESPON_TIME (0:Immediately; 1:0.01s)
J+59	-200.0000	EN_PR_GC_WATTVAR_P1 (0.1%)
J+60	0.0000	EN_PR_GC_WATTVAR_Q1 (0.1%)
J+61	-500.0000	EN_PR_GC_WATTVAR_P2 (0.1%)
J+62	0.0000	EN_PR_GC_WATTVAR_Q2 (0.1%)
J+63	-1000.0000	EN_PR_GC_WATTVAR_P3 (0.1%)
J+64	800.0000	EN_PR_GC_WATTVAR_Q3 (0.1%)
J+65	200.0000	EN_PR_GC_WATTVAR_P4 (0.1%)
J+66	0.0000	EN_PR_GC_WATTVAR_Q4 (0.1%)
J+67	500.0000	EN_PR_GC_WATTVAR_P5 (0.1%)
J+68	0.0000	EN_PR_GC_WATTVAR_Q5 (0.1%)
J+69	1000.0000	EN_PR_GC_WATTVAR_P6 (0.1%)
J+70	-800.0000	EN_PR_GC_WATTVAR_Q6 (0.1%)

4. Model Validation Benchmark Tests

4.1 Flat-run Test

No-disturbance test with Pref=1.0 p.u. and Qref=0.0 p.u. was simulated in PSS®E, as shown in Figure 4.

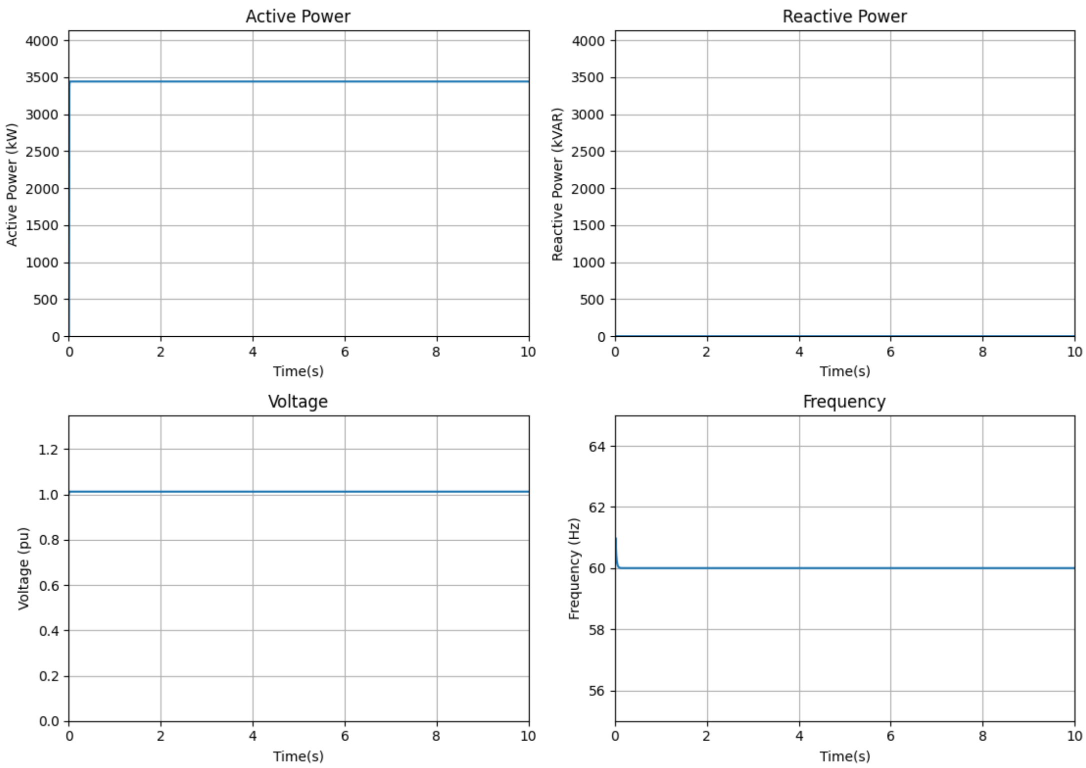

Figure 4. Dynamic plots – No fault test

4.2 Power Rising Test

Analyze the rate at which real power increases from 0 to 100% of the rated power, and conduct tests to assess the response time and performance of Delta PCS3440s, as shown in Figure 5.

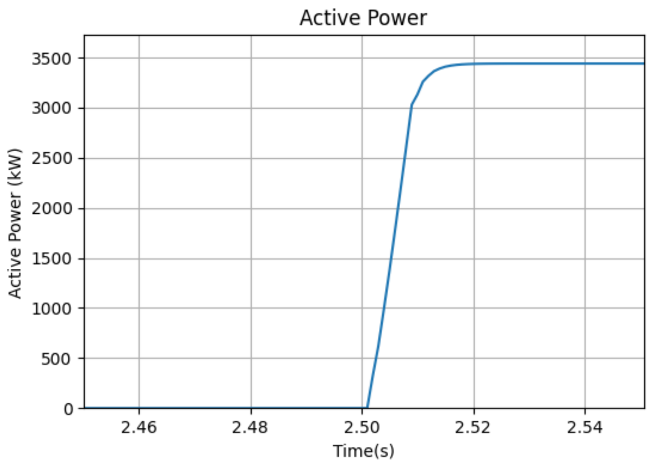

Figure 5. Dynamic plot – Power rising test

4.3 Pref Step Test

The following Pref step changes were performed in PSS®E. The step settings are shown in Table 5, where the dynamic plots are shown in Figure 6.

Table 5. Step Settings

Time [s]	Pref [p.u.]	Time [s]	Pref [p.u.]
0.0	0.0	20.0	-0.3
2.5	0.3	22.5	-0.6
5.0	0.6	25.0	-1.0
7.5	1.0	27.5	-0.8
10.0	0.8	30.0	-0.5
12.5	0.5	32.5	-0.2
15	0.2	35.0	0.0
17.5	0.0		

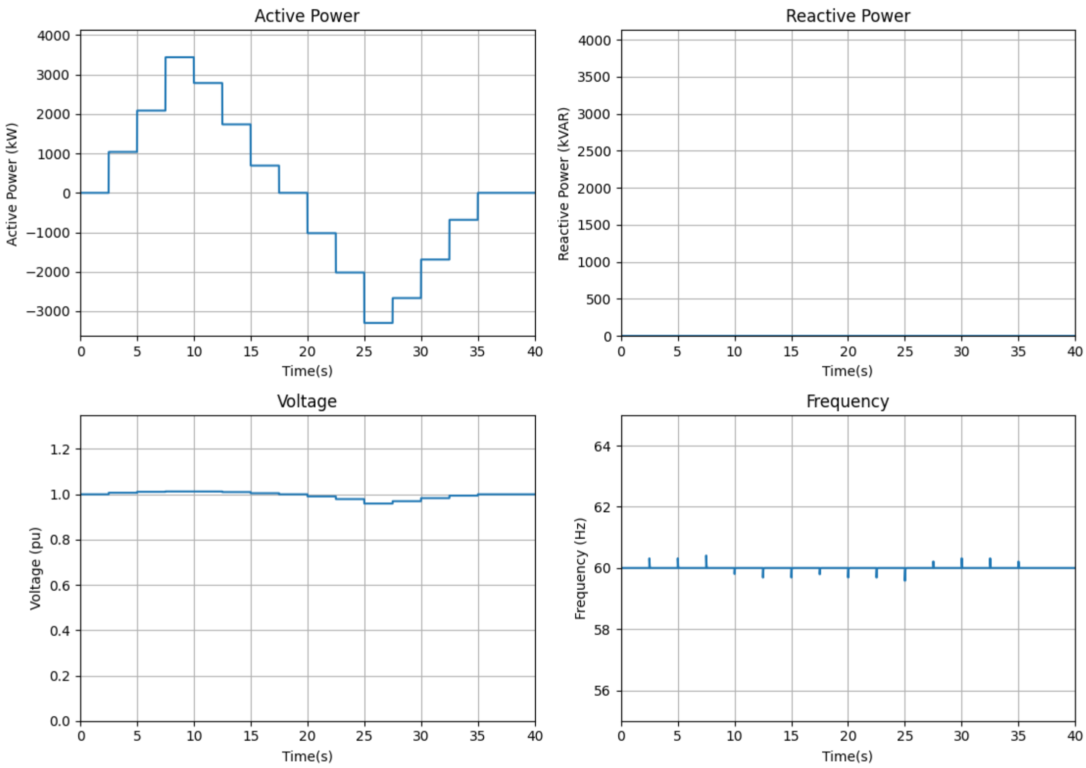

Figure 6. Dynamic plot – Pref step test

4.4 Qref Step Test

The following Qref changes were performed in PSS®E. The step settings are shown in Table 6, where the dynamic plots are shown in Figure 7.

Table 6. Step Settings

Time [s]	Qref [p.u.]
2.5	0.2
5.0	0.5
7.0	0.0
10.0	-0.3
13.0	-0.5
16.0	-0.2
20.0	-0.1

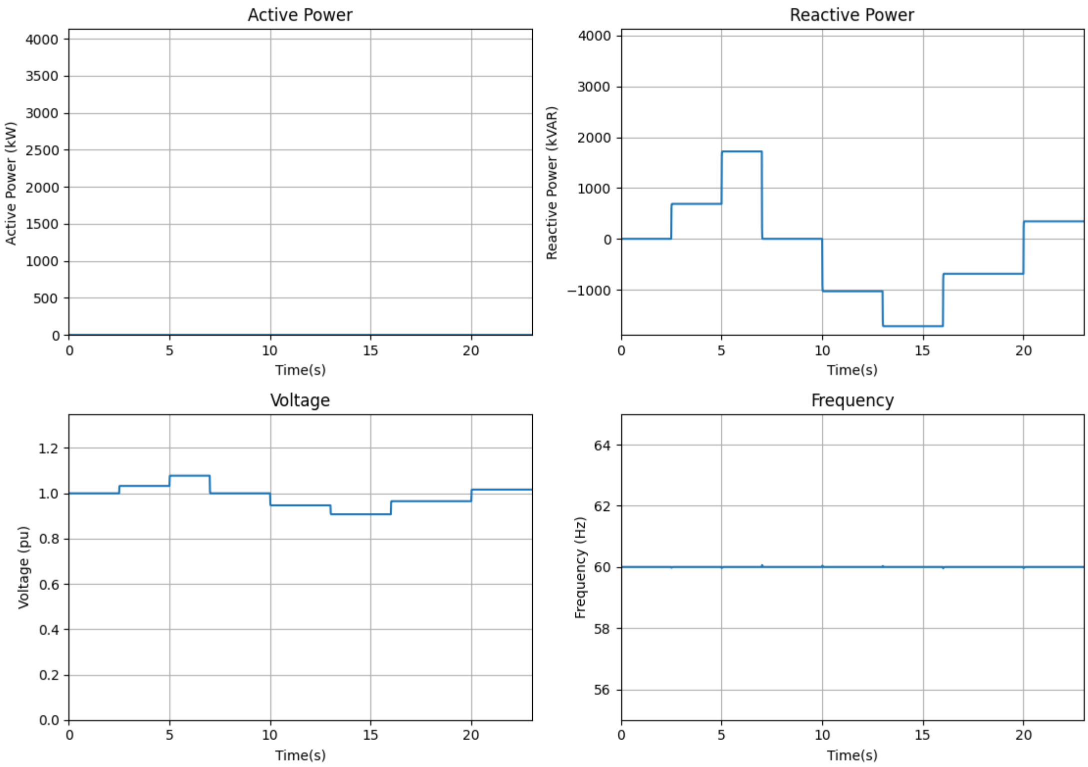

Figure 7. Dynamic plot – Qref step test

4.5 Over Voltage Protection Test

The following over voltage test was applied at the grid equivalent generator terminal to ensure the converter terminal voltage is higher than 1.15 pu for the defined duration. The over voltage condition settings are shown in Table 7, the dynamic results are shown in Figure 8.

Table 7. Over Voltage Condition Settings

Time [s]	V [p.u.]	Expected Action
0.00	1.00	Tripping
3.00	1.16	
3.50	1.16	
10.00	1.15	-

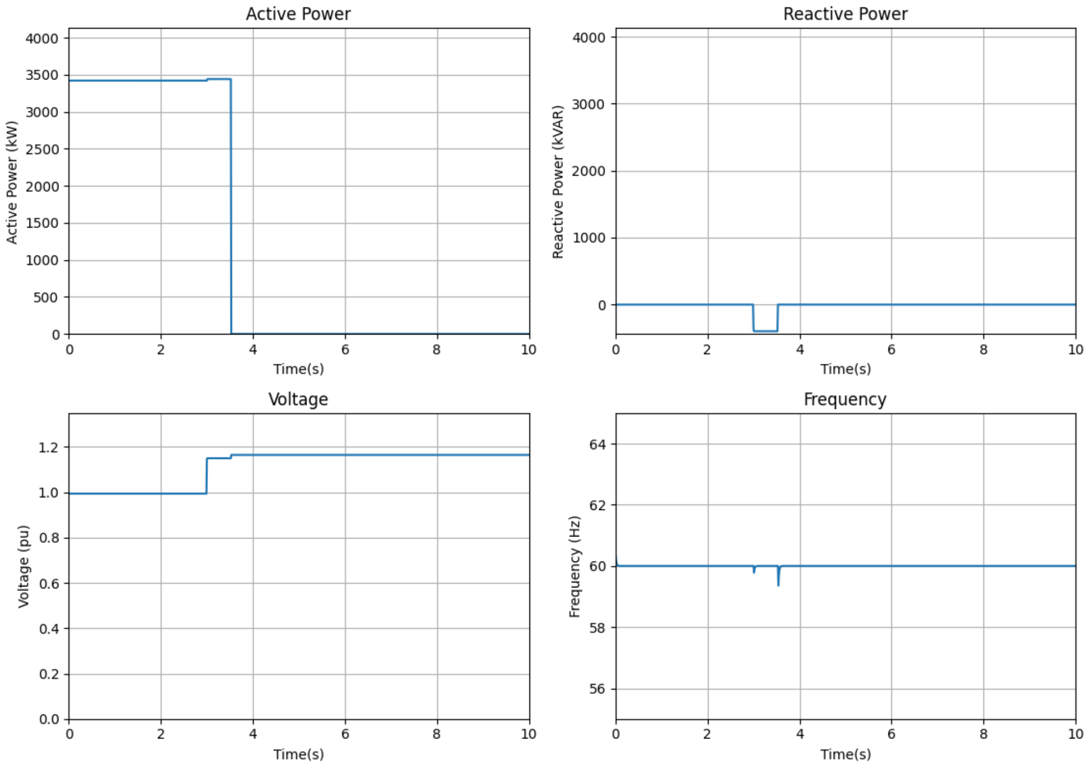

Figure 8. Dynamic plot – Over voltage test

4.6 Under Voltage Protection Test

The following under voltage test was applied at the converter terminal to ensure the converter terminal voltage is lower than 0.75 pu for the defined duration. The under voltage condition settings are shown in Table 8, the dynamic results are shown in Figure 9.

Table 8. Under Voltage Condition Settings

Time [s]	V [pu]	Expected Action
0.00	1.00	Tripping
3.00	0.70	
5.00	0.70	
10.00	0.70	-

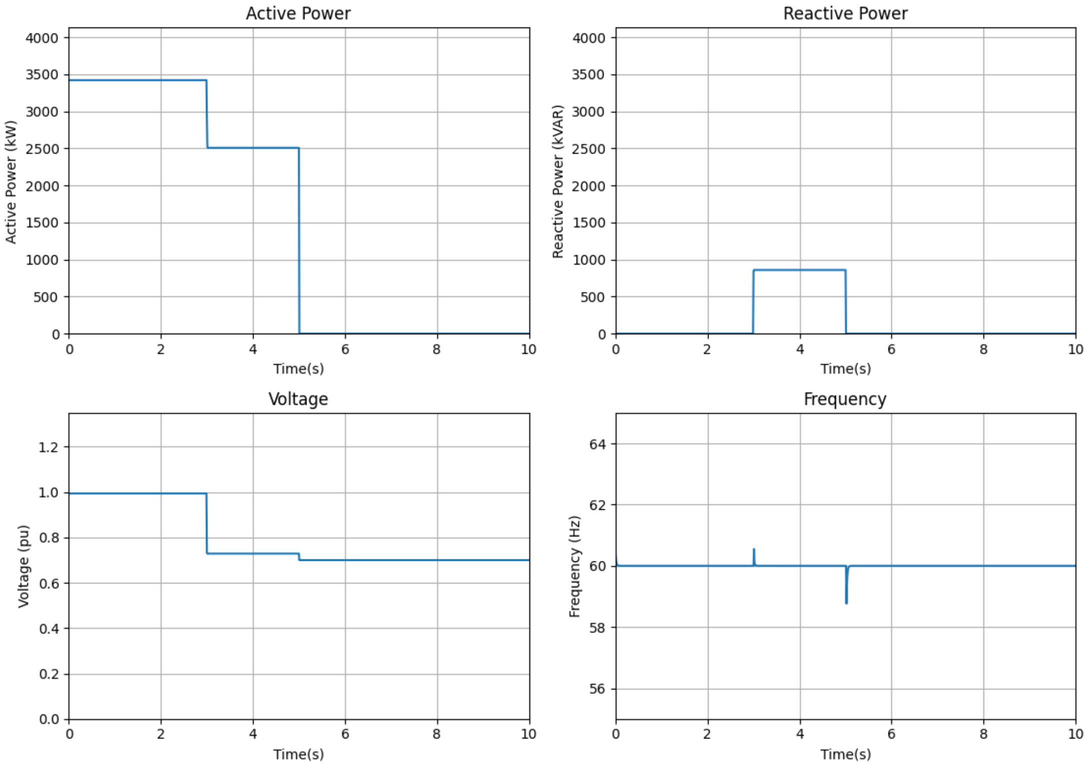

Figure 9. Dynamic plot – Under voltage test

4.7 Over Frequency Test

The following over frequency test was applied at the grid equivalent generator terminal to ensure the converter terminal bus frequency is higher than 60.6 Hz for the defined duration. The over frequency condition settings are shown in Table 9, the dynamic results are shown in Figure 10.

Table 9. Over Frequency Condition Settings

Time [s]	F [Hz]	Expected Action
0.00	60.0	Trpping
3.00	61.0	
183.0	61.0	
200.0	61.0	-

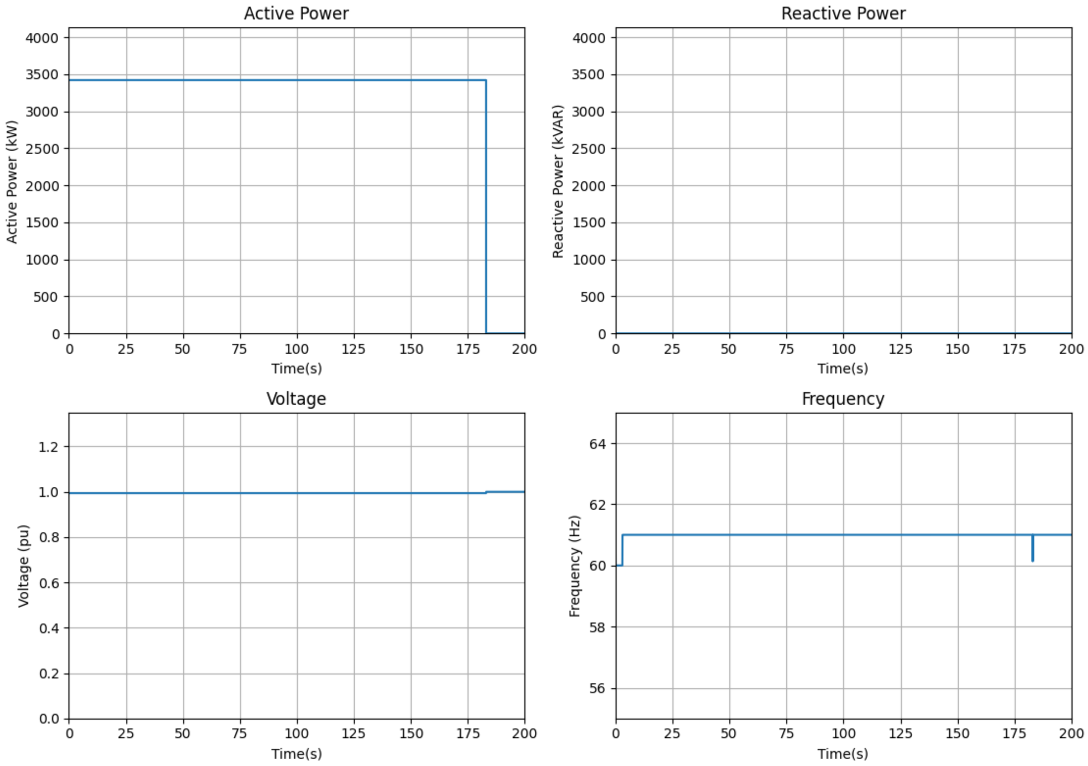

Figure 10. Dynamic plot – Over frequency test

4.8 Under Frequency Test

The following under frequency test was applied at the grid equivalent generator terminal to ensure the converter terminal bus frequency is lower than 59.4 Hz for the defined duration. The under frequency condition settings are shown in Table 10, the dynamic results are shown in Figure 11.

Table 10. Under Frequency Condition Settings

Time [s]	F [Hz]	Expected Action
0.00	60.0	Trpping
3.00	59.0	
183.0	59.0	
200.0	59.0	-

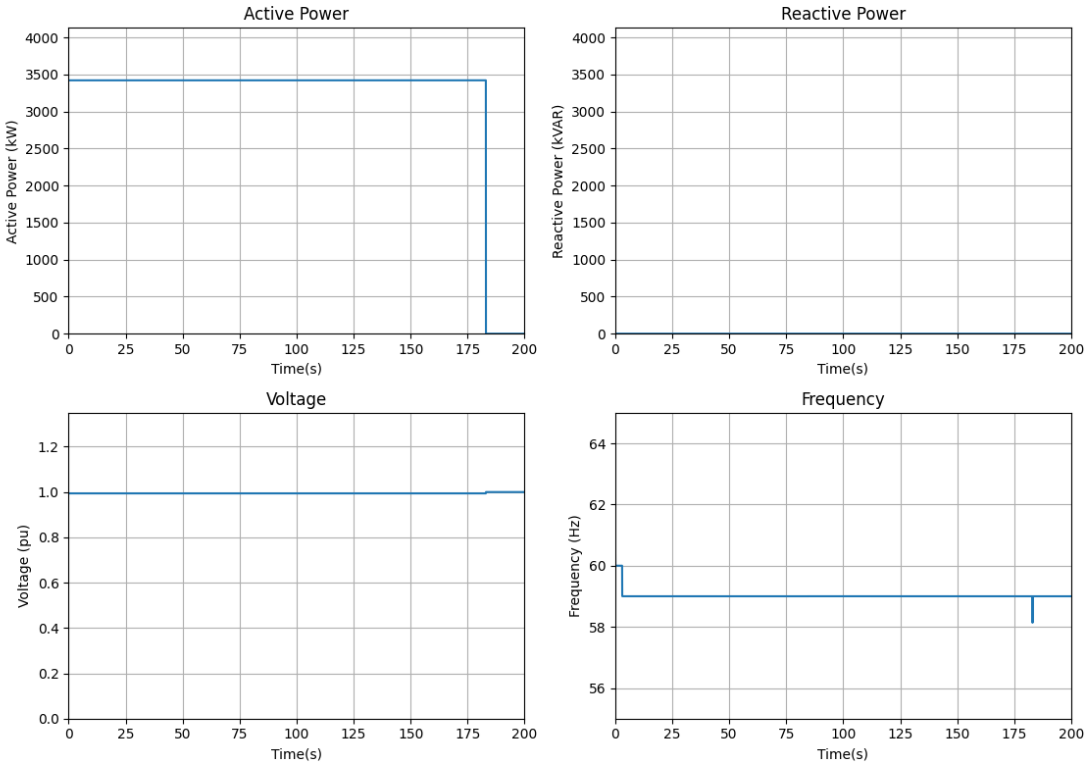

Figure 11. Dynamic plot – Under frequency test

4.9 HVRT Test

The following over voltage test was applied at the grid equivalent generator terminal to ensure the converter terminal voltage is higher than 1.10 pu for the defined duration. The over voltage condition settings are shown in Table 11, the dynamic results are shown in Figure 12.

Table 11. Over Voltage Condition Settings

Time [s]	V [pu]	Expected Action
0.00	1.00	Ride-through
3.00	1.15	
4.00	1.15	
10.00	1.15	-

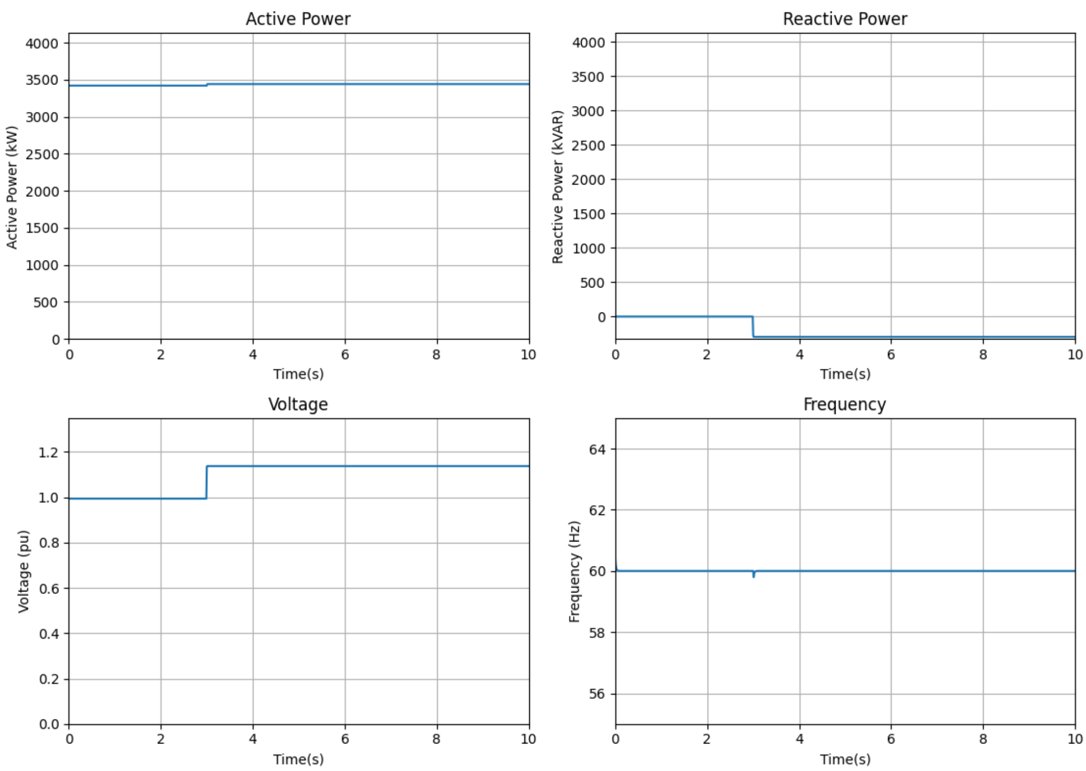

Figure 12. Dynamic plot – HVRT test

4.10 LVRT Test

The following under voltage test was applied at the grid equivalent generator terminal to ensure the converter terminal voltage is lower than 0.90 pu for the defined duration. The under voltage condition settings are shown in Table 12, the dynamic results are shown in Figure 13.

Table 12. Under Voltage Condition Settings

Time [s]	V [pu]	Expected Action
0.00	1.00	Ride-through
3.00	0.85	
9.00	0.85	
20.00	0.85	-

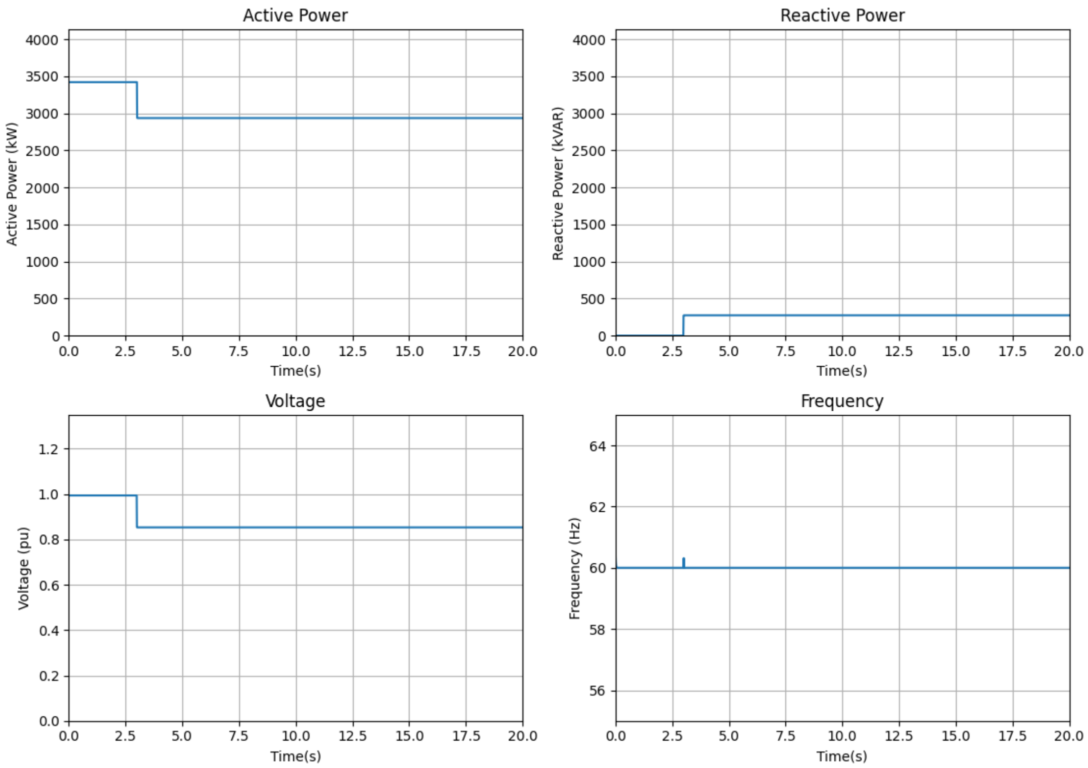

Figure 13. Dynamic plot – LVRT test

4.11 FRT (OF) Test

The following over frequency test was applied at the grid equivalent generator terminal to ensure the frequency is higher than 61.2 Hz for the defined duration. The over frequency condition settings are shown in Table 13, the dynamic results are shown in Figure 14.

Table 13. Over Frequency Condition Settings

Time [s]	F [Hz]	Expected Action
0.00	60	Ride-through
10.00	61.5	
309.0	61.5	
320.0	61.5	-

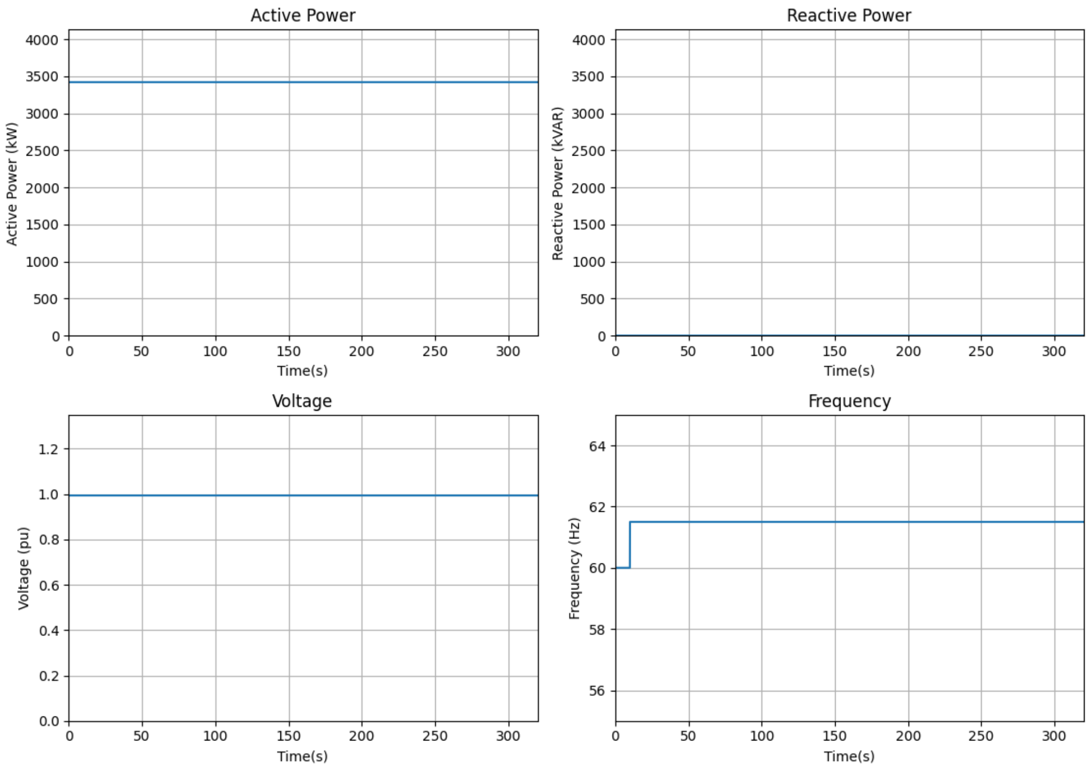

Figure 14. Dynamic plot – FRT test

4.12 FRT (UF) Test

The following under frequency test was applied at the grid equivalent generator terminal to ensure the frequency is lower than 58.8 Hz for the defined duration. The under frequnecy condition settings are shown in Table 14, the dynamic results are shown in Figure 15.

Table 14. Under frequency Condition Settings

Time [s]	F [Hz]	Expected Action
0.00	60	Ride-through
10.00	58.5	
309.0	58.5	
320.0	58.5	-

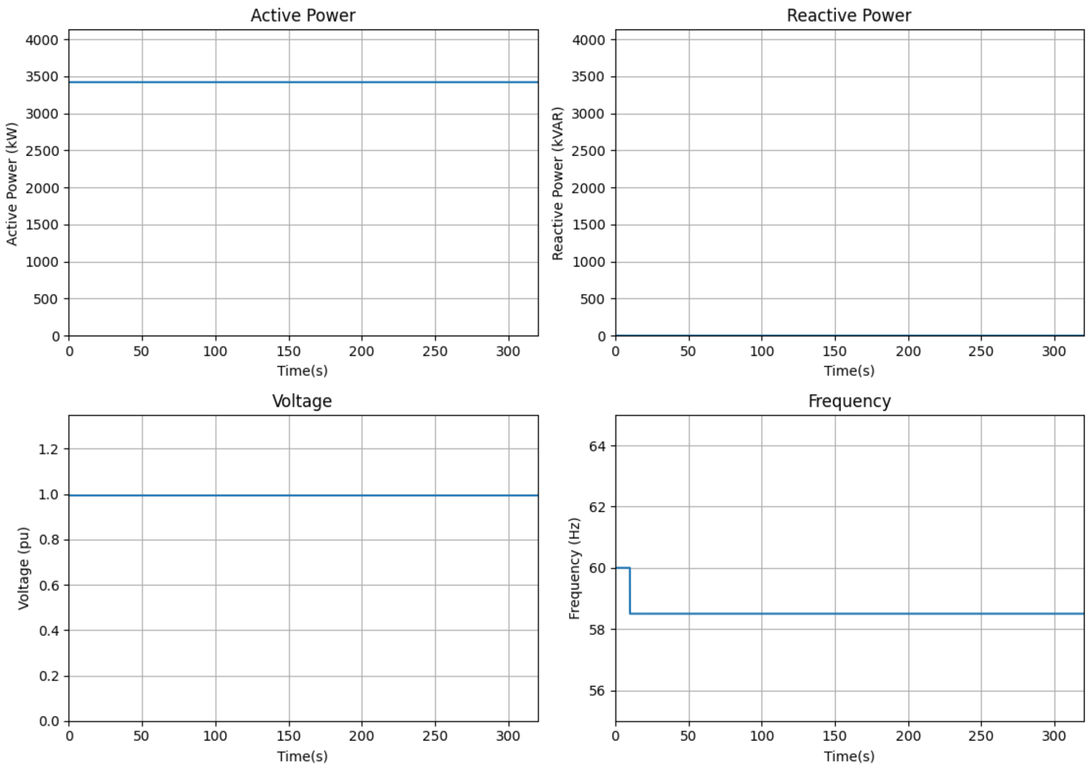

Figure 15. Dynamic plot – FRT test

5. Discussions, Conclusions and Recommendations

This report describes the PSS®E dynamic models developed by Delta Electronics, inc. for the Delta PCS3440s converter. We conducted various tests on the system based on the user-defined model, yielding insights into system response outcomes and assessing the model's reliability and stability across each test.

End of File